MIL-T-48842(PA)
9 June 1975
SUPERSEDING
See Section 6

MILITARY SPECIFICATION

TUBE, STATIC, SELF-ALIGNING, M2

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of static tube, entitled as Tube, Static, Self-Aligning, M2 which is used on the Nike Hercules warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification For.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1168	- Lot Numbering for Ammunition.

FSC 1135

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DRAWINGS

ORDNANCE CORPS

8524862	- Structure Assembly Nose Forward.
8524929	- Swivel Probe Station 6.533, Nose Forward.
8785802	- Tube, Static, Self-Aligning XM2.
8785804	- Mast.
8785807	- Nose Collar.
8785808	- Coupling Static.
8785809	- Static Probe.
8785811	- Probe Assembly Static.
8785812	- Nose Body.
8785815	- Adjusting Screw.
8804080	- Container, Metal, Amm, for Tube, Static, Self-Aligning.
8805803	- Probe Protector Assembly, H-620.
8805858	- Spider and Yoke Assembly.
8816903	- Equipment Test Panel.
EL-8815264	- Equipment List.

PUBLICATIONS

PICATINNY ARSENAL

QAI-PA-1	- Quality Assurance Instructions for Reporting Data on Nuclear Weapons.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The static tube shall comply with all requirements specified on the drawing (dwg) 8785802, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

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3.4 Serialization. A serial number shall be assigned to the item. Serial numbers required on these items shall be obtained by means of a letter addressed to: Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTENTION: SARPA-QA-T-M. The letter of request shall include:

a. Exact nomenclature of the item as specified on the applicable drawing.
b. Name of the contractor.
c. Contract number or procurement request order number.
d. Number of serial numbers required.

3.5 Leakage. The leakage rate of the static tube shall not exceed three millimeters (mm) of mercury (Hg) per minute per 10 cubic inch volume when tested as specified in 4.4.1. Leakage rate per 10 cubic inch volume _____ mm Hg Test code number 835001.

3.6 Rate of flow. The orifices in the static probe assembly, after assembly in the static tube, shall return the system to atmospheric pressure ± 1 mm Hg within 8 seconds, when tested as specified in 4.4.2. Test ___ seconds Test code number 836001.

3.7 Movement of probe.

3.7.1 Degree of movement. The tip of the probe shall have freedom of movement in all directions of 3.34 inches (6.68 inch diameter) centered about a and c of static coupling on drawing 8785802. The maximum movement shall be when the mechanical stop on the vane and body assembly contacts the aft yoke. At this point, the vane shall not contact the nose collar when tested as specified in 4.4.3.1.

3.7.2 Resistance to movement. The maximum torque to move the probe from the centerline to the 3.34-inch minimum radius shall not exceed 20 inch-ounces. The maximum torque to return the probe to the centerline shall not exceed 2.0 inch-ounces when tested as specified in 4.4.3.2. Torque required to return the probe to the centerline ___ inch ounces. Test code number 837201.

3.8 Probe balance. The static tube shall be balanced within 0 to 0.798 inch-ounces when tested as specified in 4.4.4.

3.9 Center of gravity. The center of gravity of the movable mass, less bellows shall be as specified on drawing 8785802 when tested as specified in 4.4.5.

3.10 Environmental testing (see MIL-STD-810).

3.10.1 Vibration. The static tube shall show no evidence of mechanical damage such as broken of loose parts and shall comply with the requirements of 3.5 and 3.7.2 after being tested as specified in 4.4.6.1.

3.10.2 Acceleration. The static tube shall show no evidence of mechanical damage such as broken or loose parts and shall comply with the requirements of 3.5 and 3.7.2 after being tested as specified in 4.4.6.2.

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3.10.3 Shock. The static tube shall show no evidence of mechanical damage such as broken or loose parts and shall comply with the requirements of 3.5 and 3.7.2 after being tested as specified in 4.4.6.3.

3.10.4 Life. The static tube shall be capable of withstanding 10,000 revolutions without evidence of mechanical damage or loosening of parts, and shall meet the requirements of paragraph 3.5 and 3.7.2 after being tested as specified in 4.4.6.4.

3.10.5 Temperature shock. The static tube shall show no evidence of mechanical damage or deterioration such as swelling, cracking, or loosening of parts and shall comply with the requirements of 3.5 and 3.7.2 after being tested as specified in 4.4.6.5.

3.10.6 Humidity. The static tube shall show no evidence of deterioration such as corrosion, swelling, or pitting and shall comply with the requirements of 3.5 and 3.7.2 after being tested as specified in 4.4.6.6.

3.10.7 Container leakage. The packed container shall show no evidence of leakage when tested as specified in 4.4.6.7.

3.10.8 Container vibration. The packed container shall show no evidence of fractures, chipping, damage to internal components, or loosening of the static tube when tested as specified in 4.4.6.8. After this test, the static tube shall comply with the requirements of 3.5 and 3.7.2.

3.10.9 Drop test. The packed container shall show no evidence of fractures, distortion, damage to internal components, or loosening of the static tube when tested as specified in 4.4.6.9. After this test, the static tube shall comply with the requirements of 3.5 and 3.7.2.

3.11 Coded serialization requirements (see 6.1.2). The contractor shall supply feedback data on punch cards for any item for which coding is required in section 3. The feedback data shall be in accordance with QAI-PA-1.

3.12 Workmanship.

3.12.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.12.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.12.3 Burrs. Parts shall be free from burrs.

3.12.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.12.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 3 complete assemblies (see dwg 8785802) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table I. Tests shall be conducted in order specified in Table I.

Table I

First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of defects/characteristics (see 4.3.2.1)		X
2. Leakage (see 3.5)		X
3. Rate of flow (see 3.6)		X
4. Movement of probe (see 3.7)		X
5. Probe balance (see 3.8)		X
6. Center of gravity (see 3.9)		X
7. Vibration (see 3.10.1)	X	
8. Acceleration (see 3.10.2)	X	
9. Shock (see 3.10.3)	X	
10. Life (see 3.10.4)	X	
11. Temperature shock (see 3.10.5)	X	
12. Humidity (see 3.10.6)	X	
13. Container leakage (see 3.10.7)		X
14. Container vibration (see 3.10.8)		X
15. Drop test (see 3.10.9)	X	

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4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table II and Table III), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table II
Major Defects/Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table III
Minor Defects/Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of defects/characteristics (see 3.2). Critical, major and minor defects/characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Static probe (see dwg 8785809, covering a detail of dwg 8785802).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Wall thickness, min	Gage
	102. Location of holes	Gage
	103. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.12)	Visual

4.3.2.1.2 Nose collar (see dwg 8785807 covering a detail of dwg 8785802).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Pitch diameter of large thread	Gage
	102. Minor diameter of large thread, maximum (max)	Gage
	103. Length to rear of large thread undercut	Gage

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Major: (Cont.)
104. Length to rear of large thread, min.....Gage
105. Length to large thread, max.....Gage
106. Objective evidence relating to material missing or improper (see 3.1).....Visual

Minor:
201. Evidence of poor workmanship (see 3.12).....Visual

4.3.2.1.3 Coupling, static (see dwg 8785808 covering a detail of dwg 8785802).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Pitch diameter of thread, min.	Gage
	102. Major diameter of thread, min.	Gage
	103. Length to beginning of thread, max.	Gage
	104. Length to rear of thread, min.	Gage
	105. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.12)	Visual

4.3.2.1.4 Mast (see dwg 8785804 covering a detail of dwg 8785802).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Pitch diameter of thread, max.	Gage
	102. Minor diameter of thread, max.	Gage
	103. Length to rear of thread, min.	Gage
	104. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.12)	Visual

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4.3.2.1.5 Spider and yoke assembly (see dwg 8805858 covering a detail of dwg 8785802).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Pitch diameter of external thread, min	Gage
	102. Major diameter of external thread, min	Gage
	103. Length of external thread undercut, max	Gage
	104. Adjusting screw not seated properly	Visual
	105. Any component or operation missing	Visual
	106. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.12)	Visual

4.3.2.1.6 Nose, body (see dwg 8785812 covering a detail of dwg 8785802).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Pitch diameter of large thread, max	Gage
	102. Minor diameter of large thread, max	Gage
	103. Length to beginning of thread, max	Gage
	104. Length to rear of thread, min	Gage
	105. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.12)	Visual

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4.3.2.1.7 Probe assembly, static (see dwg 8785811 covering a detail of dwg 8785802).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Braze inadequate	Visual
	102. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.12)	Visual

4.3.2.1.8 Adjusting screw (see dwg 8785815 covering a detail of dwg 8785802).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Pitch diameter of thread, min	Gage
	102. Major diameter of thread, min	Gage
	103. Length to thread, max	Gage
	104. Total length, min	Gage
	105. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.12)	Visual

4.3.2.1.9 Tube, static, self-aligning (see dwg 8785802).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Center of gravity incorrect	Test (see 4.4.5)
	102. Concentricity of nose collar at .50 location with nose collars thread and mask	Gage

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Major: (Cont.)

103.	Concentricity of vane and body assembly with static probe at .64 location	Gage
104.	Concentricity of static probe at 11.95 location with static probe at .64 location	Gage
105.	Concentricity of static probe at 12.58 location with static probe at .64 location	Gage
106.	Concentricity of fin at 5.06 location with static probe at .64 location	Gage
107.	Plane of fin not in plane with centerline of static tube	Gage
108.	Diameter of static coupling rear of groove	Gage
109.	Diameter of packing groove	Gage
110.	Width of packing groove	Gage
111.	Diameter of orifice holes	Gage
112.	Diameter at rear end of mast	Gage
113.	Diameter across wrench flat	Gage
114.	No step up between surface of nose and surface of body	Visual
115.	Burrs or foreign matter in orifice holes	Visual
116.	Finish improper	Visual
117.	Any component or operation missing	Visual
118.	Braze inadequate	Visual
119.	Marking misleading, missing or unidentifiable	Visual
120.	Serialization missing or improper (see 3.4)	Visual
121.	Objective evidence relating to material missing or improper (see 3.1)	Visual

Minor:

201.	Evidence of poor workmanship (see 3.12)	Visual

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4.3.2.1.10 Probe protector assembly prior to assembling dust cover (see dwg 8805803).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Weld inadequate	Visual
102.	Hardware missing or loose	Visual/Manual
103.	Masking missing	Visual
104.	Component missing	Visual
105.	Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
201.	Coating missing or inadequate (except for slight scratches)	Visual
202.	Marking misleading, missing or illegible	Visual
203.	Evidence of poor workmanship (see 3.12)	Visual

4.3.2.1.11 Probe protector assembly (see dwg 8805803).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Dust cover torn, cut, or seam loose	Visual
102.	Hardware missing or loose	Visual/Manual
103.	Marking missing, misleading or illegible	Visual
104.	Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
201.	Color improper or missing	Visual
202.	Evidence of poor workmanship (see 3.12)	Visual

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4.3.2.1.12 Metal container, prior to sealing (see dwg 8804080).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. End wrench missing or loose	Visual/Manual
	102. Static tube missing or improperly aligned	Visual
	103. Support center missing	Viusal
	104. Allen wrench missing or improperly bolted	Visual
	105. Packing material missing or inadequate	Visual
	106. Packing list missing	Viusal
	107. Dessicant missing or insufficient	Visual
	108. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.12)	Visual

4.3.2.1.13 Container metal (see dwg 8804080).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Container damaged	Visual
	102. Car seal missing or improperly assembled	Visual
	103. Cover loose	Visual/Manual
	104. Marking misleading or unidentifiable	Visual
	105. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
	201. Coating inadequate (except for slight scratches)	Visual
	202. Evidence of poor workmanship (see 3.12)	Visual

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4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test Classification
a. Leakage	Critical
b. Rate of flow	Major
c. Movement of probe	Critical
d. Probe balance	Major
e. Container leakage	Minor
f. Vibration	Major
g. Acceleration	Major
h. Temperature shock	Major
i. Humidity	Major

Of these tests, f, g, h and i are considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Leakage, rate of flow, movement of probe, probe balance and container leakage. Each static tube in the lot shall be tested in accordance with 4.4.1 through 4.4.4 and 4.4.6.7. If any item fails to comply with 3.5 through 3.8 and 3.10.7, the item shall be rejected and removed from the lot.

4.3.3.2 Vibration, acceleration, temperature shock and humidity. A sample shall be selected at random in accordance with Table IV. The sample shall be tested in accordance with 4.4.6.1, 4.4.6.2, 4.4.6.5 and 4.4.6.6. If any item fails to comply with 3.10.1, 3.10.2, 3.10.5 and 3.10.6, the lot shall be rejected.

Table IV

Lot size	Sample size
2-8	1
9-15	2
16-25	3
26-40	5
41-65	8
66-110	13
111-180	22
180-300	36
301 and over	MIL-STD-105 (AQL 0.25 percent)

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4.3.4 Acceptance inspection equipment.

4.3.4.1 Major defects/characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8815264). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor defects/characteristics and tests. Acceptance inspection equipment for minor defects/characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Leakage. The orifices in the probe shall be temporarily sealed. The static coupling shall be joined to a pressure sensing device and a source of vacuum that is capable of obtaining 500 mm Hg negative pressure. A positive shut off valve shall be located between the pressure sensing device and the source of vacuum. When the temperature of the static tube and testing equipment has stabilized (see 6.4) at ambient temperature and the pressure within the static tube and testing equipment has stabilized at 500 mm Hg negative pressure, the positive shut off valve shall be closed. The pressure sensing device shall be observed for 10 minutes and the rate of leakage recorded. During the test, the probe shall be rotated 10 times at its maximum limit of travel.

4.4.2 Rate of flow. The orifices in the probe shall be temporarily sealed. The static coupling shall be connected to a pressure sensing device, then connected to a chamber of 400 + 20 cubic inches volume, and finally attached to a source of 520 + 25 mm Hg negative pressure. A positive shut off valve shall be located between the chamber and the source of negative pressure. When the temperature of the static tube and testing equipment has stabilized at ambient temperature and the pressure within the static tube and chamber has stabilized

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at 520 + 25 mm Hg negative pressure, the positive shut off valve shall be closed. The seal will then be removed from the probe orifices, and the time required for the system to return to atmospheric pressure shall be recorded.

4.4.3 Movement of probe.

4.4.3.1 Degree of movement. The static tube shall be mounted in a fixture which simulates the mounted characteristics of the missile. The fixture shall establish a zero centerline at the tip of the probe within 0.010 inch of the centerline of the mounting device. A reference line at 3.340 + 0.010 inches from and parallel to the zero centerline shall be established. The probe tip shall pass the 3.340 mark at a minimum of four points in line with the vanes of the assembly. The probe shall then be moved beyond the minimum point until the mechanical stop hits. The vanes shall not contact the nose collar. If the probe tip does not reach the minimum point, or the vane contacts the nose collar, the probe shall be rejected.

4.4.3.2 Resistance to movement. With the static tube mounted in the fixture referred to in paragraph 4.4.3.1, the probe tip shall be moved to the minimum displacement line. The torque shall not exceed 20 inch-ounces (567 inch-grams). The probe shall then be moved to the zero point and the torque shall not exceed 2.0 inch-ounces (57 inch-grams). The displacement from the zero point shall be in four directions approximately midway between the vanes. Any assembly that exceeds the allowable torque shall be rejected. Record the torque values.

4.4.4 Probe balance. The static tube shall be mounted in a fixture which simulates the mounting characteristics of the missile. The fixture shall establish zero centerline at the tip of the probe within 0.010 inch of the centerline of the mounting device. The fixture shall also establish a line that will indicate when the probe is displaced 12.5° upward from the horizontal. This 12.5° shall be computed using mean dimensions of all components. The assembly shall be held vertical and the probe manually aligned to the zero mark on the fixture. The probe shall be displaced to the 12.5° mark and the torque required to so displace it, recorded (R1). Place the fixture in a horizontal position and again deflect the probe to the 12.5° mark and record the torque (R2). After completing values R1 and R2, the assembly shall be rotated 90° and the test shall be repeated to obtain values R1 and R3. The difference between any two of the four readings taken shall be within 0 to 0.798 inch ounces.

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4.4.5 Center of gravity. The center of gravity of the movable mass of the static tube, less bellows, shall be determined by any moment method that is approved by the contracting officer. The movable mass of the static tube, less bellows, shall consist of the following components, probe with forward end fitting, vane and body assembly, two nose set screws, forward yoke, two forward yoke set screws, spring pin, two bearings and two adjustable screws.

4.4.6 Environmental testing (see 3.10).

4.4.6.1 Vibration. The static tube shall be mounted in a fixture simulating actual mounting conditions (see dwg 8524862 and dwg 8524929). A constant aerodynamic stabilizing force shall be simulated throughout the test by inserting a sponge rubber collar so as to contact the aft inner surface of the vane and body assembly and the aft yoke. The static tube plus fixture shall be vibrated through a frequency range of 5 to 500 to 5 Hertz (Hz) in 15-minute cycles for four cycles. From 5 to 55 Hz the test shall be at a double amplitude of 0.02 inch. From 55 to 500 Hz the test shall be at a constant acceleration of five gravity units (g). The vibrations shall be measured on the nose collar. After this test the static tube shall be tested in accordance with 4.4.1 and 4.4.3.2.

4.4.6.2 Acceleration. The static tube shall be mounted in a fixture simulating actual mounting conditions (see dwg 8524862 amd dwg 8524929). A constant aerodynamic stabilizing force shall be simulated throughout the test by inserting a sponge rubber collar so as to contact the aft inner surface of the vane and body assembly and the aft yoke. The static tube plus fixture shall be subjected to a constant acceleration of 40 g for ten seconds minimum in both directions of the longitudinal axis, and 25 g for ten seconds in both directions of each of the other two mutually perpendicular axes. After the acceleration test, the static tube shall be tested in accordance with 4.4.1 and 4.4.3.2.

4.4.6.3 Shock. The static tube shall be mounted in a fixture simulating actual mounting conditions (see dwg 8524862 and 8524929). A constant aerodynamic stabilizing force shall be simulated throughout the test by inserting a sponge rubber collar so as to contact the aft inner surface of the vane and body assembly and the aft yoke. The static tube plus fixture shall first be subjected to 18 shocks of 25 g as follows: Three shocks in each direction of each of the three mutually perpendicular axes. The static tube plus fixture shall then be mounted vertically and subjected to three shocks of 50 g in such a manner that the shock impulses travel from the static coupling end of the tube toward the tip probe. All of the 21 shocks shall have a total duration of 11 $\pm$ 1 milliseconds. After this test the static tube shall be tested as specified in 4.4.1 and 4.4.3.2.

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4.4.6.4 Life. The static tube shall be held in a fixture which maintains 10 degrees deflection, but which allows the assembly to rotate like a universal joint. The static tube shall then be revolved at 30 rotations per minute for 6 hours. After this test, the static tube shall be tested in accordance with 4.4.1 and 4.4.3.2.

4.4.6.5 Temperature shock. The static tube shall be conditioned at -65° ± 3°F until the temperature of the tube becomes stabilized. The static tube shall then be removed from the cold chamber and within one minute placed in a hot chamber having a temperature 600° ± 5°F and held therein for 120 ± 20 seconds. Within one hour after removal from the hot chamber, the static tube shall be tested in accordance with 4.4.1 and 4.4.3.2.

4.4.6.6 Humidity. The static tube shall be tested in accordance with Specification MIL-E-5272, paragraph 4.4.1, Procedure I, with a H-620 probe protector installed (see dwg 8805803). After the humidity test the static tube shall be tested in accordance with 4.4.1 and 4.4.3.2.

4.4.6.7 Container leakage. The leakage test shall be conducted in accordance with the equipment test panel in dwg 8816903.

4.4.6.8 Container vibration. The packed container shall be rigidly mounted to a suitable vibration producing apparatus. The package shall be vibrated through the range of frequency from 10 to 55 to 10 Hz at a constant double amplitude of 0.060 ± 0.006 inch for a duration of one hour in each of three mutually perpendicular major axes. Each cycle, 10 to 55 to 10 Hz shall have a duration of between 55 to 65 seconds. The frequency shall be varied uniformly throughout the frequency range specified. The static tube shall then be removed from the container and tested as specified in 4.4.1 and 4.4.3.2.

4.4.6.9 Drop test. The static tube used in the vibration test (see 4.4.6.2) shall be repacked, using the same container and the same packing material. The package shall be permitted to fall freely ten (10) times from a height of 36 inches onto a solid concrete surface of a minimum thickness of 12 inches. The 36-inch drop height shall be the shortest distance from the concrete surface to the nearest point of the container. There shall be no ropes or other suspended media attached to the container during the fall. One drop shall be made on each of four (4) simulated diagonally opposite corners for a total of 8 drops. One drop shall be made on each of two adjacent side walls (90° travel) for a total of two (2) drops. When the diagonal chime drops are being conducted, the package shall be suspended with the

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center of gravity vertically above the point of impact. When the sidewall drops are being conducted, the sidewall shall be positioned in a place parallel to the concrete slab. After completion of all tests, the static tube shall be removed from the container and tested in accordance 4.4.1 and 4.4.3.2.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The static tube shall be packaged in accordance with dwg 8804088 using the specified moisture barrier.

5.1.2 Level C. The static tube shall be packaged in accordance with dwg 8804088 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The static tube shall be packed in accordance with dwg 8804088.

5.3 Marking. Marking shall be as specified in accordance with dwg 8804088.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1 and 3.10). Data specified in 3.5, 3.6, 3.7.2 and 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Supersession data. This specification supersedes the requirements of Purchase Description X-PA-PD-1026 dated 8 October 1959.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1135-A207

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PENALTY FOR PRIVATE USE $300

COMMANDER
Picatinny Arsenal
ATTN: SARPA-QA-N-Q
Dover, New Jersey 07801

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. *Fold on lines on reverse side, staple in corner, and send* to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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